

VENABU, Norway

WMO: 013800

Lat: 61.651N

Lon: 10.109E

Elev: 930

StdP: 90.64

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-19.9	-25.4	0.4	-21.7	-22.9	0.5	-19.5	11.2	-4.1	9.9	-4.3	1.4	0	0.757

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	22.0	13.0	19.9	12.3	17.7	11.3	14.3	19.5	13.3	17.6	12.3	16.3	2.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.3	10.0	15.2	11.3	9.3	14.6	10.4	8.8	13.6	43.0	20.0	40.1	17.5	37.5	16.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.8	6.6	-25.3	24.1	2.2	2.3	-26.8	25.7	-28.1	27.1	-29.3	28.3	-30.9	30.0		
(5)			-25.5	15.3	2.1	1.2	-27.1	16.1	-28.3	16.8	-29.5	17.5	-31.1	18.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.5	-9.3	-8.3	-5.2	-0.6	4.6	8.9	11.9	10.0	6.1	0.3	-5.0	-7.6		
	DBStd	8.38	4.93	5.09	4.54	3.91	4.54	3.29	3.25	2.55	2.97	3.34	4.47	5.13		
	HDD10.0	3597	599	513	471	318	177	58	16	32	119	299	449	545		
	HDD18.3	6493	857	746	729	568	426	283	199	259	366	558	699	803		
	CDD10.0	146	0	0	0	0	11	26	76	31	3	0	0	0		
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH23.3	19	0	0	0	0	1	2	17	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	2.8	2.6	2.7	3.2	2.8	3.1	3.2	2.8	2.5	2.8	2.7	2.4	2.6		
(15) Precipitation	PrecAvg	606	41	26	22	28	53	71	91	89	52	50	46	35		
	PrecMax	820	73	57	51	63	133	144	156	143	102	161	143	73		
	PrecMin	478	6	7	0	3	6	0	39	17	14	6	7	5		
	PrecStd	83	19	13	14	16	30	35	32	35	22	34	28	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	3.7	7.8	12.7	21.5	22.9	25.3	20.2	17.4	11.5	5.6	3.8		
		MCWB	0.8	0.8	3.1	6.2	11.3	12.2	14.8	13.3	11.0	9.3	4.4	1.9		
	2%	DB	0.5	1.9	5.2	10.0	18.8	20.5	23.3	18.4	14.9	8.6	3.5	2.2		
		MCWB	-0.7	-0.3	1.9	4.4	10.3	11.4	14.1	12.1	10.0	5.7	2.6	0.8		
	5%	DB	-1.0	0.2	3.4	7.6	15.5	17.6	21.3	16.7	12.8	6.8	2.2	0.5		
		MCWB	-1.6	-1.2	0.6	3.2	8.7	10.2	13.3	11.5	9.2	4.7	1.4	-0.3		
	10%	DB	-2.5	-1.6	1.2	5.6	12.6	15.3	18.8	14.9	11.1	5.3	0.7	-0.9		
		MCWB	-3.1	-2.5	-1.0	2.1	7.8	9.2	12.4	10.8	8.4	3.8	0.1	-1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	1.5	3.7	6.8	12.3	13.4	16.0	14.6	12.6	10.1	4.6	2.3		
		MCDB	2.0	3.3	6.6	12.0	18.7	18.5	22.7	18.3	14.3	10.6	5.4	3.3		
	2%	WB	-0.4	0.0	2.2	4.8	11.0	12.4	14.9	13.4	11.0	6.7	2.9	1.0		
		MCDB	0.5	1.3	5.2	9.0	17.3	17.6	20.9	16.2	13.3	7.9	3.6	1.9		
	5%	WB	-1.6	-1.2	0.7	3.5	9.5	11.4	14.0	12.5	9.8	5.2	1.6	-0.2		
		MCDB	-0.9	0.1	2.9	7.0	14.6	16.4	19.8	15.2	12.0	6.3	2.1	0.5		
	10%	WB	-3.1	-2.4	-0.5	2.4	7.9	10.2	13.0	11.4	8.6	3.9	0.2	-1.5		
		MCDB	-2.6	-1.5	1.1	5.4	12.0	14.6	17.8	14.1	10.8	5.1	0.6	-0.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.9	8.3	8.9	9.0	9.3	9.5	8.5	7.5	6.1	5.4	6.1		
		MCDBR	6.6	8.3	9.8	10.9	14.0	13.5	13.7	11.2	10.8	8.8	5.6	6.4		
		MCWBR	6.3	6.7	6.8	6.5	7.0	6.3	5.9	5.9	6.5	6.3	5.0	6.0		
		MCDBR	6.5	7.5	8.9	10.0	12.4	11.3	11.8	9.2	8.9	7.0	5.3	6.5		
	5% WB	MCWBR	6.2	6.1	6.3	6.0	6.7	5.7	5.6	5.4	5.9	5.7	4.8	6.1		
(40) Clear-Sky Solar Irradiance	taub		0.192	0.239	0.269	0.301	0.312	0.310	0.322	0.314	0.285	0.268	0.205	0.159		
	taud		2.124	2.277	2.273	2.287	2.453	2.524	2.493	2.507	2.575	2.499	2.182	1.974		
	Ebn at Noon		569	766	858	893	913	920	896	876	850	734	539	417		
	Edh at Noon		33	59	86	105	97	92	93	84	65	49	31	21		
(44) All-Sky Solar Radiation	RadAvg		0.28	0.95	2.51	4.32	4.89	5.13	4.71	3.64	2.34	1.06	0.35	0.15		
	RadStd		0.04	0.17	0.24	0.40	0.46	0.48	0.55	0.31	0.26	0.16	0.06	0.01		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page